



Prepare Better. Assess Smarter.

Purpose-built digital assessment and learning technology
for aviation training institutions across India.

*Covering every stage of the assessment lifecycle — from question creation
and examination delivery through performance intelligence and student learning.*

DGCA-Style Assessment

Performance Analytics

Student Learning

Instructor Intelligence

THE CORE CONCEPT

What if your cadets had already taken the DGCA exam — before the actual exam?

Examination day should measure knowledge — not familiarity with unfamiliar software. Students encountering a new examination interface for the first time during a live assessment must simultaneously navigate an unknown environment, manage a countdown, and answer difficult questions under pressure.

Repeated exposure to a realistic examination environment builds familiarity. Familiarity reduces cognitive friction. ZYXEN allows cadets to practice in an assessment environment designed to closely resemble the structure and operational flow of DGCA-style computer-based examinations.

"The student should be thinking about the question — not figuring out the examination interface."

Central design principle, ZYXEN Assessment Platform

FIRST EXPOSURE

01 Initial encounter with the examination environment

**PRACTICE**

02 Repeated assessments across all examination subjects

**FAMILIARITY**

03 The interface becomes instinctive, not unfamiliar

**CONFIDENCE**

04 Full cognitive focus shifts to answering the question

**READINESS**

05 The cadet is prepared — not just for the content, but the experience

The ZYXEN Familiarity Framework



CBT ASSESSMENT PLATFORM

An Examination Experience Designed Around Aviation.

ZYXEN is not a generic quiz platform adapted for aviation. It is purpose-built from the ground up for the assessment and examination preparation requirements of aviation training institutions, flight academies and ground schools preparing cadets for DGCA-style computer-based examinations.

Every element — from the examination interface to the question structure, timing mechanics and result delivery — is designed around how aviation examinations actually work, not around how a generic assessment product works.

REALISTIC EXAMINATION EXPERIENCE

A structured CBT environment designed to closely mirror the DGCA-style examination flow — including navigation, question palette, review mechanics and structured question delivery.

FLEXIBLE ASSESSMENT

Create subject-specific examinations, timed mock tests, practice sessions and scheduled assessments for individual cadets or a batch of cadets — at any stage of their training.

INSTRUCTOR CONTROL

Faculty maintain full control over question banks, examination creation, batch management, scheduling and the delivery of assessments across subjects and student cohorts.

STUDENT PERFORMANCE INTELLIGENCE

Turn examination results into structured academic insight. Subject, topic and question-level analysis makes every assessment an opportunity to understand the student better.

ASSESSMENT INTEGRITY

Built-in examination controls support supervised, structured assessments with appropriate monitoring, access controls and session management for serious examination conduct.

SUBJECTS COVERED

Air Navigation

Aviation Meteorology

Air Regulations

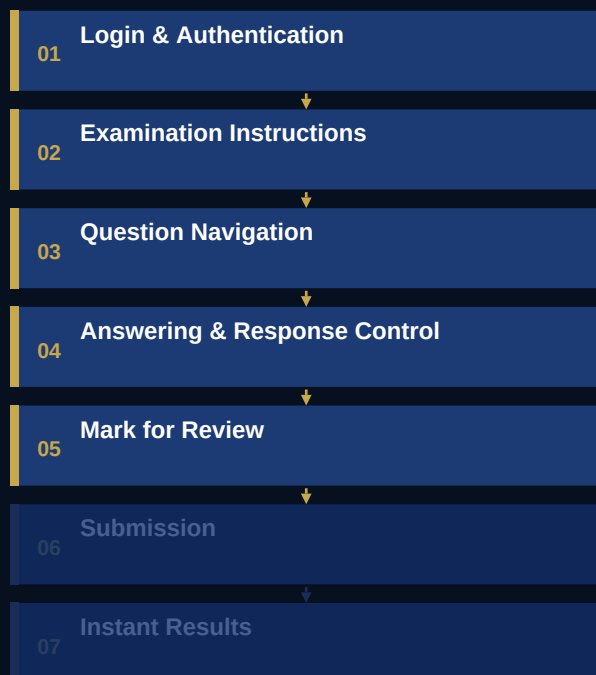
Technical General

Technical Specific

EXAMINATION EXPERIENCE

Make the Environment Familiar Before the Stakes Are Real.

Every element of the ZYXEN examination environment is designed to feel familiar before a cadet faces a real assessment. When the examination begins, they should already know exactly where to find the timer, how to navigate between questions, and how to mark items for review.



ZYXEN Assessment Platform

Air Navigation — Mock Examination 03
02:45:30

Question 14 of 60
14 answered · 2 marked · 44 remaining

Which of the following best describes the relationship between TAS and IAS as altitude increases in a standard atmosphere?

☐ A. TAS increases; IAS remains relatively constant

☒ B. TAS increases with altitude; IAS decreases

☐ C. Both TAS and IAS increase proportionally with altitude

☐ D. TAS remains constant; IAS increases as altitude increases

← Previous
Mark for Review
Next →



PERFORMANCE INTELLIGENCE

More Than a Score.

Every ZYXEN assessment generates structured performance data — not just a final mark. Instructors can move beyond asking whether a student passed and begin asking where that student needs the most support.

"Instead of asking only Did the student pass? — instructors can begin asking Where does this student need help?"

Subject-Wise Performance

Understand results at subject level across all assessments

Topic-Level Insights

Drill into specific topic performance within each subject

Individual Progress

Track each student's improvement and trajectory over time

Post-Exam Feedback

Detailed explanations and solutions delivered after each test

EXAMINATION RESULT

Input

**SUBJECT PERFORMANCE****TOPIC INSIGHT****WEAKNESS IDENTIFIED****TARGETED PRACTICE****Batch Comparison****REASSESSMENT**

Surface areas requiring focused revision and practice

Assessment → Intelligence Cascade

Examination Readiness

Monitor readiness across a batch ahead of real assessments

Solution Explanations

Step-by-step answer walkthroughs for every question

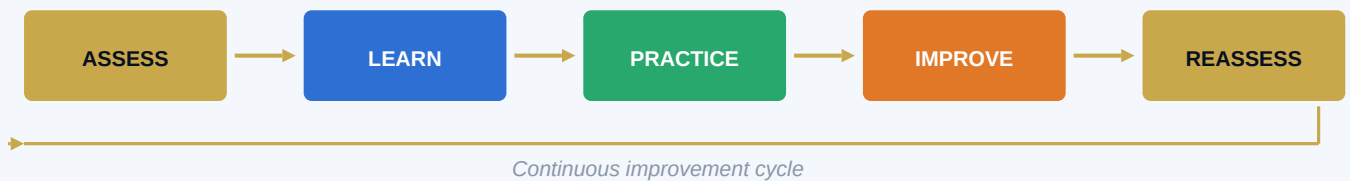
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STUDENT LEARNING ECOSYSTEM

From Assessment to Continuous Learning.

ZYXEN extends beyond examination delivery. The learning ecosystem connects assessment outcomes directly to structured learning activity — creating a natural loop of assessment, review, practice, improvement and reassessment.



CURRENT CAPABILITIES

- **Structured Learning Materials**
Course content organised by subject and module
- **Assignment Delivery**
Distribute assignments to student batches
- **Student Progress Tracking**
Monitor individual progress through course material
- **Practice Modules**
Allow students to practice independently anytime
- **Academic Resource Management**
Organise and distribute study materials to students
- **Instructor-Led Content Support**
Faculty can manage and update learning content
- **Course Content Organisation**
Structure learning material systematically by subject
- **Batch & Subject Organisation**
Assign content to specific batches and cohorts
- **Assessment-Linked Learning**
Connect examination results to relevant study content
- **Downloadable Study Resources**
Students can access reference material for offline study
- **Faculty Content Management**
Instructors manage what content students can access
- **Progress Reporting**
Structured reports on student engagement and progress

ON THE PRODUCT ROADMAP

Live classroom integration · Recorded learning modules · AI-assisted study support · Personalised learning paths

These capabilities form part of the ZYXEN development roadmap and are not currently deployed.



STAKEHOLDER VIEW

Give Instructors Visibility. Give Cadets Direction.

ZYXEN serves every stakeholder in an aviation training environment — from the cadet sitting an examination to the academy director monitoring cohort readiness.

FOR INSTRUCTORS

Faculty & Ground School Staff

- **Manage & Schedule Assessments**
Create, configure and schedule examinations for any batch at any time.
- **Question Bank Management**
Organise questions by subject, topic and difficulty across all DGCA subjects.
- **Monitor Student Performance**
Track individual and batch performance across every conducted assessment.
- **Identify Learning Gaps**
Surface the specific topics and areas where students are underperforming.
- **Compare Batch Performance**
Benchmark multiple student cohorts against each other on identical assessments.
- **Review Assessment Outcomes**
Access structured, detailed reports for every examination conducted.
- **Track Progress Over Time**
Monitor each student's improvement from first attempt through examination.

FOR ACADEMY LEADERSHIP

Directors, HOT & Academic Management

- **Overall Academic Performance**
Understand examination outcomes across all subjects, batches and time periods.
- **Batch Readiness Monitoring**
Assess whether a cohort is on track for DGCA examination preparedness.
- **Identify Intervention Areas**
Flag subjects or student groups that require additional instruction or support.
- **Centralised Assessment View**
A single, structured view of all assessment activity across the institution.
- **Structured Reporting**
Performance data presented in a clear, accessible format for leadership review.
- **Examination Activity Overview**
Monitor which assessments are being conducted, when, and by which batches.
- **Progress Benchmarking**
Track institutional performance trends and identify improvement over time.

QUESTION BANK & ASSESSMENT

Build, Organise and Deliver Better Assessments.

ZYXEN provides a structured environment for building, managing and delivering assessments across subjects, batches and examination schedules — with support for rich question content including diagrams and mathematical notation.

- **Centralised Question Bank**
All questions organised by subject, topic and question type
- **Subject & Topic Organisation**
Clean taxonomy aligned to DGCA examination subject structure
- **Rich Question Content**
Support for text, diagrams and mathematical content types
- **Examination Builder**
Configure assessments from curated question bank selections
- **Batch Assignment**
Assign examinations to specific student batches and groups
- **Scheduled Examination Windows**
Define start and end times for controlled assessment delivery
- **Practice Assessment Mode**
Students can attempt practice papers at any time independently
- **Controlled Assessment Workflows**
Structured processes from creation through to result delivery

01 QUESTION BANK
Questions organised by subject & topic



02 ASSESSMENT BUILDER
Configure examinations from question sets



03 BATCH ASSIGNMENT
Assign to the relevant student cohort



04 EXAMINATION
Conducted under controlled conditions



05 ANALYTICS
Results converted into performance insight

One structured workflow — from question creation through to performance insight.

ASSESSMENT INTEGRITY & RELIABILITY

Designed for Serious Assessment.

Aviation examinations demand rigor. ZYXEN is built to help academies conduct assessments with confidence — supporting examination integrity, controlled access, structured session management and reliable result delivery.

"Designed to help academies conduct assessments with confidence."

Examination Integrity Controls

Measures designed to maintain the integrity of supervised assessments during live examination sessions, supporting proper conduct throughout.

Controlled Access & Session Management

Structured access workflows ensure only authorised students can commence and complete assigned assessments within defined parameters.

Assessment Monitoring

Supervisors have appropriate visibility into active examination sessions to support and maintain proper examination conduct standards.

Reliable Submission & Result Delivery

Submission workflows and result delivery are designed for accuracy, consistency and structured access by students and instructors.

Connectivity Resilience

Assessment sessions are designed to handle connectivity interruptions appropriately, protecting the integrity of submitted responses.

Structured Examination Scheduling

Examinations are administered within defined time windows with appropriate access controls at the start and close of each session.

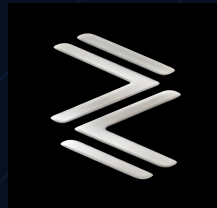
PLATFORM VISION & ROADMAP

Building the Digital Assessment & Learning Infrastructure for Aviation Training.

ZYXEN is not building an examination website. It is building a long-term digital learning ecosystem — a purpose-built infrastructure for aviation training institutions that grows with the needs of their students, faculty and examination requirements.



Roadmap capabilities are planned future developments and are not currently deployed.



ZYXEN

Aviation Assessment & Learning Platform

See ZYXEN in Action.

The best way to understand the platform is to experience it firsthand.

15-Minute Walkthrough

15-minute focused demonstration of the platform in action.

Complimentary Trial

Experience ZYXEN directly without any commitment.

Academy-Tailored Demo

Demonstrations configured to your institution's context.

Request a Demonstration

zyxen.in

contact@zyxen.in

No commitment required · Complimentary for aviation training institutions